MIT

MIVA OPEN UNIVERSITY

# School of Post Graduate Studies

Master of Information Technology (MIT)

**2024/2025 Academic Session**

MIT 8212 – Seminar: Industry Applications and Management in Information Technology

Credit Units: 3

## 1. Course Overview

This seminar course offers students the opportunity to investigate, analyze, and present practical applications of Information Technology (IT) management within industry contexts. It emphasizes critical engagement with **real-world IT problems,** application of **industry best practices,** and the development of **research-informed, industry-relevant presentations.**

The seminar fosters both **academic depth** and **practical exposure** by combining case studies, reflective learning, and student-led projects that demonstrate mastery of IT management principles in organizational and industrial settings.

## 2. Learning Outcomes

By the end of this course, students will be able to:

1. Analyze current industry trends and their implications for IT management.
2. Apply data-driven decision-making techniques to practical industry problems.
3. Investigate real-world IT management challenges using systematic investigative techniques.
4. Demonstrate proficiency in IT infrastructure planning, cost-benefit analysis, and IT management tools.
5. Collaborate across disciplines to address cross-functional organizational challenges.
6. Present findings from real-world projects in a structured, evidence-based seminar.
7. Communicate effectively with industry stakeholders through reports and presentations.
8. Engage in reflective learning to connect theory with practice.

The structure of the course is divided into **five main sections:**

- Batch 1: Industry Landscape and Emerging IT Trends.
- Batch 2: Data-Driven Decision-Making and Investigative Approaches.
- Batch 3: Solving Industry Problems through IT Management.
- Batch 4: Cross-Functional Collaboration and Project Execution.
- Batch 5: Communication, Evaluation, and Reflective Learning.

Your major output will be a **20–25 page seminar paper** and a **presentation** based on your group's chosen topic.

## Batch 1: Industry Landscape and Emerging IT Trends

1. **Analyzing Global and Local IT Trends in Agriculture**
   - Research Question: How are digital platforms (mobile apps, IoT, drones) transforming agricultural productivity in Nigeria?

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2. **Blockchain and Cloud Adoption in Health IT**
   - Research Question: What are the challenges and opportunities of using blockchain and cloud services for secure patient record management?
3. **Emerging Technologies in Logistics and Supply Chain**
   - Research Question: How can IoT-based tracking reduce inefficiencies and losses in last-mile delivery?
4. **Digital Transformation in Sports Management**
   - Research Question: What role does AI-based analytics play in improving player performance and fan engagement?
5. **Trends in Education Technology (EdTech)**
   - Research Question: What barriers do Nigerian universities face in adopting AI-driven learning platforms?

## Batch 2: Data-Driven Decision-Making and Investigative Approaches

6. **Data-Driven Decision-Making in Financial Services**
   - Research Question: How can machine learning models improve fraud detection in digital banking?
7. **Investigative Techniques in Public Health IT Systems**
   - Research Question: What investigative methods can improve the accuracy of vaccination data in developing countries?
8. **Big Data Analytics in Smart Cities (Government Sector)**
   - Research Question: How can traffic and energy usage data be leveraged for sustainable urban planning?
9. **AI-Enabled Predictive Analytics in Sports**
   - Research Question: Can predictive models reduce athlete injuries and improve training schedules?
10. **Data Governance and Ethics in Higher Education Institutions**
    - Research Question: How can universities ensure data privacy and compliance in student management systems?

## Batch 3: Solving Industry Problems through IT Management

11. **Case Studies of Failed ERP Implementations in Manufacturing**
    - Research Question: What lessons can Nigerian manufacturers learn from ERP project failures?
12. **IT Infrastructure Planning in Banking**
    - Research Question: How can banks plan resilient IT infrastructure to mitigate cyberattacks?
13. **Cost-Benefit Analysis of E-Government Projects in Nigeria**
    - Research Question: Do the benefits of e-government portals outweigh the costs in terms of efficiency and public service delivery?

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14. **Smart Grid IT Management in Energy Sector**
    - Research Question: How can IT systems optimize power distribution and reduce energy losses?
15. **Evaluating IT Service Management in Aviation**
    - Research Question: How can ITIL frameworks enhance customer service in airport operations?

## Batch 4: Cross-Functional Collaboration and Project Execution

16. **Cross-Functional Collaboration in Agricultural Value Chains**
    - Research Question: How can IT platforms improve communication between farmers, markets, and logistics providers?
17. **Collaboration Between IT and Clinicians in Hospital Systems**
    - Research Question: How can IT managers and clinicians overcome resistance to electronic medical records?
18. **IT and Operations Collaboration in Logistics**
    - Research Question: How can cross-functional teams leverage IT to reduce supply chain bottlenecks?
19. **Agile Project Execution in FinTech Startups**
    - Research Question: What challenges do Nigerian FinTech startups face in adopting agile methodologies?
20. **IT Management in Sports Events**
    - Research Question: How can IT systems enhance coordination in global sports competitions (ticketing, broadcasting, and security)?

## Batch 5: Communication, Evaluation, and Reflective Learning in IT Management

21. **Evaluation and Reporting of IT in E-Learning Systems**
    - Research Question: How effective are IT evaluation metrics (KPIs, ROI) in measuring online university success?
22. **Communicating IT Value to Stakeholders in NGOs**
    - Research Question: How can NGOs better communicate the value of IT systems to donors and field staff?
23. **Using IT Management Tools in Health Projects**
    - Research Question: How do IT management tools like Jira or Tableau improve project tracking in healthcare?
24. **Stakeholder Communication in Agricultural Digital Platforms**
    - Research Question: How can IT managers design farmer-friendly communication strategies to increase adoption of digital platforms?
25. **Reflective Learning from Smart City IT Projects**
    - Research Question: What reflective lessons can be drawn from failed or delayed smart city initiatives?

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## Seminar Paper Structure

Each student's presentation/paper (20–25 pages) should include:

1. **Introduction** – Introduce the subject matter and define the industry problem being addressed. Identify a sector-specific IT management problem (e.g., poor hospital data sharing, unreliable farm market systems).
2. **Literature and Industry Review** – Summarize existing research and practice. Review industry cases/literature to show existing solutions.
3. **Investigative Approach** – Explain data collection, frameworks, or tools applied.
4. **Findings/Analysis** – Present solutions or lessons learned from case/application.
5. **Conclusions/Recommendations** – Conclude with practical insights for IT management.

**You will be graded in this course using this rubric:**

| Criteria | Description | Marks |
|---|---|---|
| **1. Introduction & Problem Definition** | Background clearly explained, problem well-defined, and research question(s) precise and industry-relevant. | 15 |
| **2. Literature & Industry Review** | Comprehensive and up-to-date coverage of academic and industry sources, critical analysis (not descriptive), and inclusion of global/Nigerian trends. | 20 |
| **3. Investigative Approach / Methodology** | Appropriate framework selected (e.g., SWOT, TAM, data analysis), clearly described steps, and justified choice of method. | 15 |
| **4. Findings & Analysis** | Depth of analysis with evidence (data/cases), linked back to literature, demonstrates originality and provides meaningful interpretation. | 25 |
| **5. Conclusions & Recommendations** | Clear conclusions, practical and actionable recommendations, and contributions to industry practice. | 15 |
| **6. Academic Writing & Integrity** | Logical structure (5-chapter flow), clarity of language, correct APA referencing, originality/plagiarism-free. | 10 |
| **Total** | | 100 |

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